DS 301 Homework 5 Solutions

Instructions: Homework is to be submitted on Canvas by the deadline stated above. Please clearly print your name and student ID number on your HW.

Show your work (including calculations) to receive full credit. Please work hard to make your submission as readable as you possibly can - **this means no raw `R` output or code in your solutions.** (unless it is asked for specifically or needed for clarity).

Code should be submitted with your homework *as a separate file* (for example, a .R file, text file, Word file, or .Rmd are all acceptable). You should mark sections of the code that correspond to different homework problems using comments (e.g. `##### Problem 1 #####`).

Problem 1: Best subset selection

The data for this problem comes from a study by Stamey et al. (1989). They examined the relationship between the level of prostate-specific antigen and a number of clinical measures in men who were about to receive a radical prostatectomy. The variables are log cancer volume (`lcavol`), log prostate weight (`lweight`), `age`, log of the amount of benign prostatic hyperplasia (`lbph`), seminal vesicle invasion (`svi`), log of capsular penetration (`lcp`), Gleason score (`gleason`), and percent of Gleason scores 4 or 5 (`ppp45`). The last column corresponds to which observations were used in the training set and which were used in the test set (`train`).

Read in the prostate data set using the following code:

```
prostate = read.table('.../prostate.data',header=TRUE)
```

In place of `'...'`, specify the pathway where you saved the dataset.

Our response of interest here is the log prostate-specific antigen (`lpsa`). We will use this data set to practice 3 common subset selection approaches.

a. **Approach 1**: Perform best subset selection on the entire data set with `lpsa` as the response. For each model size, you will obtain a 'best' model (size here is just the number of predictors in the model): M_1 is the best model with 1 predictor (size 1), M_2 is the best model with 2 predictors (size 2), and so on. Create a table of the AIC, BIC, adjusted R^2 and Mallow's C_p for each model size. Report the model with the smallest AIC, smallest BIC, largest adjusted R^2 and smallest Mallow's C_p. Do they lead to different results? Using your own judgement, choose a final model.

```
 library(leaps)
regfit = regsubsets(lpsa~.-train,data=prostate,nbest=1,nvmax=8)
regfit.sum = summary(regfit)
regfit.sum

n = dim(prostate)[1]
```

```
p = rowSums(regfit.sum$which)
adjr2 = regfit.sum$adjr2
cp = regfit.sum$cp
rss = regfit.sum$rss
AIC = n*log(rss/n) + 2*(p)
BIC = n*log(rss/n) + (p)*log(n)

cbind(p,rss,adjr2,cp,AIC,BIC)

  p      rss     adjr2        cp       AIC       BIC
1 2 58.91478 0.5345839 27.406210 -44.36603 -39.21661
2 3 51.74218 0.5868977 14.747299 -54.95846 -47.23433
3 4 46.56844 0.6242063  6.173546 -63.17744 -52.87859
4 5 45.59547 0.6280585  6.185065 -63.22555 -50.35199
5 6 44.43668 0.6335279  5.816805 -63.72263 -48.27437
6 7 43.77597 0.6349654  6.466493 -63.17571 -45.15273
7 8 43.10756 0.6365002  7.100428 -62.66823 -42.07054
8 9 43.05842 0.6327886  9.000000 -60.77886 -37.60646
```

Note: your C_p will look slightly different if you allowed the predictor ***train*** *to be in your model. This is because C_p uses the full model to estimate σ^2 and whether or not* ***train*** *is in your model will affect this estimate. Either way is fine since we don't care about the absolute value of C_p but its magnitude in comparison with other models.*

The model with the smallest AIC and C_p is M_5:

```
(Intercept)      lcavol     lweight         age        lbph         svi
      0.495       0.544       0.588      -0.016       0.101       0.715
```

The model with the smallest BIC is M_3:

```
(Intercept)      lcavol     lweight         svi
     -0.777       0.526       0.662       0.666
```

The model with the largest adjusted R^2 is M_7:

```
(Intercept)      lcavol     lweight         age        lbph         svi         lcp
      0.494       0.570       0.614      -0.021       0.097       0.752      -0.105
```

The criteria, in general, lead you to pick different final models. It's really up to students what they would pick. As long as they communicate their rationale (i.e. AIC leads to more complex model versus BIC leads to simpler but consistent model), students can receive full credit.

b. **Approach 2**: The dataset has already been split into a training and test set. Construct your training and test set based on this split. You may use the following code for convenience:

```
 train = subset(prostate,train==TRUE)[,1:9]
test = subset(prostate,train==FALSE)[,1:9]
```

For each model size, you will obtain a 'best' model. Fit each of those models on the training set. Then evaluate the model performance on the test set by computing their test MSE. Choose a final model based on prediction accuracy. Fit that model to the full dataset and report your final model here.

My test MSEs for each 'best' model are:

```
        M1        M2        M3        M4        M5        M6        M7        M8
0.4797387 0.4924823 0.4005308 0.4563321 0.4859242 0.5485933 0.5165135 0.5212740
```

Model M_3 has the smallest test MSE; refitting it to the full dataset gives me the final model:

```
  > coef(lm(lpsa~lcavol+lweight+svi,data=prostate))
(Intercept)      lcavol     lweight         svi
 -0.7771566   0.5258519   0.6617699   0.6656666
```

c. **Approach 3**: This approach is used to select the optimal **size**, not which predictors will end up in our model. Split the dataset into k folds (you decide what k should be). We will perform best subset selection within each of the k training sets. Here are more detailed instructions:

   i. For each fold $k = 1, \ldots, K$:

      1. Perform best subset selection using all the data except for those in fold k (training set). For each model size, you will obtain a 'best' model.
      2. For each 'best' model, evaluate the test MSE on the data in fold k (test set).
      3. Store the test MSE for each model.

      Once you have completed this for all k folds, take the average of your test MSEs for each model size. In other words, for all k models of size 1, you will compute their k-fold cross-validated error. For all the k models of size 2, you will compute their k-fold cross-validated errors, and so on. Report your 8 CV errors here.

      I set $k = 5$.

```
k = 5
folds = sample(1:k,nrow(prostate),replace=TRUE)

val.errors = matrix(NA,k,8)

## loop over k folds (for j in 1:k)
for(j in 1:k){
  test = prostate[folds==j,]
  train = prostate[folds!=j,]

  best.fit = regsubsets(lpsa~.,data=train,nbest=1,nvmax=8)

  for(i in 1:8){
```

```
    test.mat = model.matrix(lpsa~.,data=test)

    coef.m = coef(best.fit,id=i)

    pred = test.mat[,names(coef.m)]%*%coef.m
    val.errors[j,i] = mean((test$lpsa-pred)^2)
  }

}
```

My 5-fold cross-validated errors for each models are as follow:

```
    1     2     3     4     5     6     7     8
0.648 0.615 0.549 0.573 0.576 0.589 0.561 0.573
```

ii. Choose the model size that gives you the smallest CV error. Now perform best subset selection on the full data set again in order to obtain this final model. Report that model here. (For example, suppose cross-validation selected a 5-predictor model. I would perform best subset selection on the full data set again in order to obtain the 5-predictor model.)

The model of size 3 has the smallest 5-fold CV errors. Refitting on the full dataset gives me the following final model with 3 predictors:

```
  coef(regfit,3)
(Intercept)      lcavol     lweight         svi
 -0.7771566   0.5258519   0.6617699   0.6656666
```

Note that this is the same as the model BIC gives us in Approach 1.

Problem 2: Concept Review

(a) For the following statement, state whether or not it is True or False. **Briefly justify your answer**.
Suppose I have 3 models to pick from:

$$M_A : Y \sim X_1 + X_2 + X_3 + X_4 + X_5$$
$$M_B : Y \sim X_6 + X_7 + X_8 + X_9 + X_{10}$$
$$M_C : Y \sim X_1 + X_2 + X_7 + X_9 + X_{10}$$

Using AIC, BIC, Mallow's C_p, adjusted R^2 could lead us to pick different final models.

False. Since the models are of the same size, the criteria just boil down to RSS (the penalty term is a constant across models).

(b) For the following statement, state whether or not it is True or False. **Briefly justify your answer**.
Suppose I have two models:

$$M_3 : Y \sim X_1 + X_2 + X_4$$
$$M_4 : Y \sim X_4 + X_5 + X_6 + X_7$$

It must be that $\text{RSS}_3 \geq \text{RSS}_4$.

False. This is only true with nested models (when the smaller model contains an exact subset of predictors in the larger model).

(c) For the following statement, state whether or not it is True or False. **Briefly justify your answer**.
Suppose I have two models:

$$M_2 : Y \sim X_4 + X_5$$
$$M_4 : Y \sim X_4 + X_5 + X_6 + X_7$$

It must be that $\text{RSS}_2 \geq \text{RSS}_4$.

True. These are nested models.

(d) What advantages are there to using AIC/BIC instead of using the test MSE as our model selection criteria? Explain.

The main advantage is that AIC/BIC can be thought of as indirect estimates of our test MSE. This means that in implementation we could utilize the full dataset to perform model selection. This can be very advantageous if our sample size n is small.

Problem 3: Diagnostics for MLR

(a) List the four assumptions we make when fitting a multiple linear regression model.

We assume $E(\epsilon) = 0$, the errors are uncorrelated, we have constant variance (σ^2), and that there is an approximately linear relationship between the response Y and predictors X.

(b) True or False? In order to fit a multiple linear regression model (i.e. obtain least square estimates), we must have distributional assumptions on the random error term ϵ_i. *Briefly justify your answer.*

False. We do not need any distributional assumption on ϵ_i in order to minimize our residual sum of squares.

(c) True or False? The `lm()` output automatically gives you p-value for individual hypothesis tests: $H_0 : \beta_j = 0$ versus $H_1 : \beta_j \neq 0$, $j = 1, \ldots, p$. These hypothesis tests assume that the random error term are normally distributed. *Briefly justify your answer.*

True. `lm()` assumes that the random error term are normally distributed. Since σ^2 is estimated, we use the t-distribution and the p-values are directly computed from the t-distribution (with $n - (p + 1)$ degrees of freedom).

(d) Suppose we want to run some visual diagnostics to check whether or not the linearity assumption holds. Besides the fact that it is is time consuming, explain why it is not sufficient to check every pairwise scatterplot between Y and X_j, $j = 1, \ldots, p$.

Pairwise scatterplots between Y and X_j, $j = 1, \ldots, p$ do not take into account how the relationship between predictors affect their association with Y.

(e) To check whether or not the linearity assumption holds, we can run some diagnostics. Let's use the `Auto` dataset as an example. Run the following code:

```
m1 = lm(mpg~horsepower,data=Auto)
summary(m1)
par(mfrow=c(2,2))
plot(m1)
```

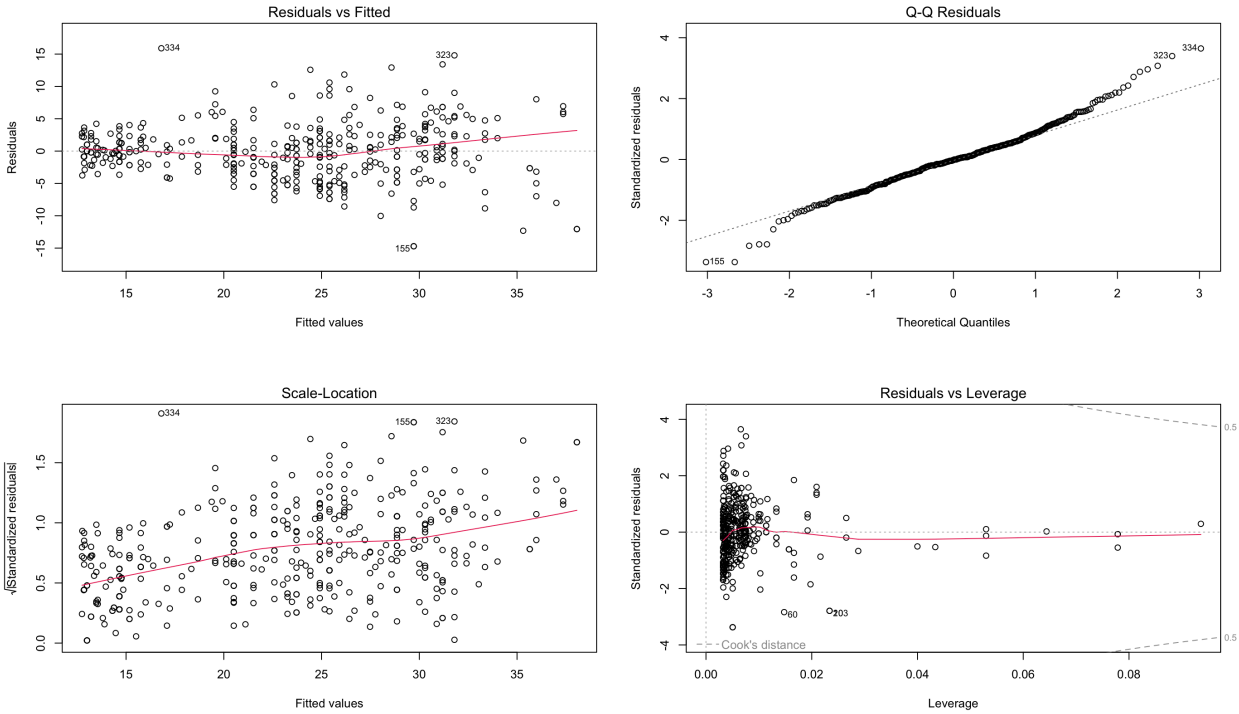
Take a look at the plot in the top left corner (titled Residuals vs Fitted). This is what we call a *residual plot* because it is literally a plot of residuals versus fitted values ($\hat{y}_i$). The red line is a smooth fit to the residuals, which is displayed in order to make it easier to identify any trends. You should observe that the plot exhibits a clear U-shape, which provides a strong indication of non-linearity in the data.

Ideally what we want to see is **no discernible pattern** in the residual plot and a relatively flat red line. A residual plot exhibiting no pattern and a flat red line indicates that the linearity assumption for our model holds. Propose a more flexible model than `m1` that might be able to accommodate some non-linearity. Present your proposed model (call it `m2`) and its corresponding residual plot here. Is it an improvement over `m1`?

Note we'll discuss the remaining plots from `plot(m1)` and their utility in an upcoming lecture.

I propose a polynomial model with degree 2:

```
m2 = lm(mpg~poly(horsepower,2),data=Auto)
par(mfrow=c(2,2))
plot(m2)
```


Residuals vs Fitted
Residuals
Fitted values
334
323
155
Q-Q Residuals
Standardized residuals
Theoretical Quantiles
323
334
155
Scale-Location
√|Standardized residuals|
Fitted values
334
155
323
Residuals vs Leverage
Standardized residuals
Leverage
Cook's distance
0.5
0.5
60
203